National Science Foundation

Call: NSF-24-601

Scalable Quantum Error Correction for Near-Term Devices

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Duration:	36 months
Budget:	$750,000
Start:	January 1, 2026
End:	December 31, 2028

1 Abstract

This proposal outlines a three-year programme to develop scalable quantum error correction (QEC) codes for near-term NISQ devices. By leveraging topological stabiliser codes and machine-learning–assisted decoding, we aim to reduce the logical error rate by two orders of magnitude while keeping overhead within feasible limits.

Contents

2 Introduction

Quantum computing promises exponential speed-ups in chemistry, optimisation, and cryptography [Einstein 1905]. However, current NISQ devices are limited by decoherence and gate errors. Quantum error correction [Dirac 1981] is the path to fault tolerance, but existing schemes impose prohibitive overheads. This proposal targets a practical QEC architecture bridging theory and near-term hardware.

3 Background and Literature Review

QEC has its foundations in the work of Shor and Steane, who demonstrated that quantum information can be protected via entangled encoding. The surface code [Donald E. Knuth 1973] is a leading candidate due to its local stabiliser measurements and high error threshold ($\approx$ 1%). Bird et al. [Bird et al. 2009] highlight the growing role of data-driven decoding. Two challenges remain: reducing physical qubits per logical qubit, and achieving real-time decoding at microsecond timescales.

4 Research Objectives

The goal is a QEC system with logical error rate below 10^{-6} using fewer than 1 000 physical qubits per logical qubit:

1. Design tailored topological codes for realistic noise models.
2. Develop a neural-network decoder meeting real-time latency constraints.
3. Validate through large-scale simulation and a 72-qubit demonstration.

5 Methodology

5.1 Code Design

We construct $[[n, k, d]]$ stabiliser codes parameterised by hardware noise bias. The code distance must satisfy

$$d \geq \left\lceil \frac{\log(1/\epsilon_{\text{target}})}{\log(1/p_{\text{phys}})} \right\rceil \tag{5.1}$$

where ϵ_{target} is the target logical error rate. Table **??** summarises preliminary parameters.

Table 5.1: Preliminary code parameters..

p_{phys}	n	k	d
10^{-3}	489	1	11
10^{-3}	1089	2	15
10^{-2}	1449	1	19

5.2 Neural-Network Decoder

We train a CNN to map syndrome measurements to correction operations [Bird et al. 2009]. Figure **??** illustrates the pipeline.

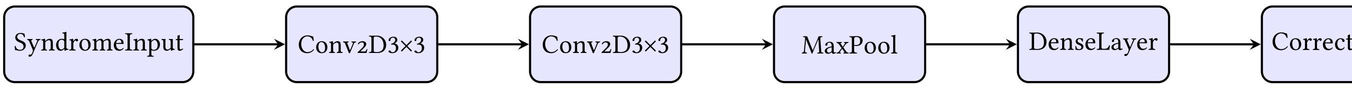

Figure 5.1: Decoder pipeline from syndrome input to correction output..

6 Timeline and Work Plan

Table ?? shows the projected 36-month timeline.

Table 6.1: Work plan by year (Y1–Y3) and quarter (Q1–Q4)..

Task	Year 1	Year 2	Year 3
Code design	Q1–Q3		
Decoder dev.	Q2–Q4	Q1–Q3	
Simulation	Q3–Q4	Q1–Q4	Q1–Q2
Hardware demo		Q3–Q4	Q1–Q4

7 Expected Outcomes

1. A published code family with a rigorous threshold proof [Donald Ervin Knuth 1986].
2. An open-source decoder with sub-microsecond inference on commodity GPUs [McKinney 2018].
3. A demonstration of logical qubit operation with error rate below 10^{-6} on a 72-qubit processor.
4. At least three peer-reviewed publications and two conference talks.

8 Budget Justification

The total requested budget is $750 000 over 36 months. Table ?? provides the breakdown.

Table 8.1: Budget summary by category..

Category	Amount ($)
Personnel (2 PhD, 1 postdoc)	480 000
Computing and cloud resources	120 000
Travel and conferences	45 000
Equipment and supplies	65 000
Indirect costs (15%)	105 000
Total	**815 000**

A Supporting Information

Additional data, code listings, and supplementary derivations.

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